

Candidate Intelligence Engine - Technical Design

Candidate: Preethi S | **Email:** prepreethi2611@gmail.com

1. Pipeline Breakdown

Detect (File type & URL) → Extract (Regex entities) → Normalize (E.164, YYYY-MM, ISO-3166) → Merge (Priority resolution) → Confidence (Per-field scoring) → Project (Runtime config) → Validate (Pydantic schema)

2. Canonical Output Schema & Normalized Formats

```
{
  "full_name": "Preethi S",
  "emails": ["prepreethi2611@gmail.com"], // lowercase
  "phones": ["+918778524631"], // E.164
  "skills": ["Java", "React"], // canonical aliases
  "location": {"city": "Coimbatore", "region": "Tamil Nadu", "country": "IN"}, // ISO-3166
  "overall_confidence": 0.825,
  "field_provenance": {
    "full_name": {"source": "resume", "confidence": 0.85, "normalized": true}
  }
}
```

3. Merge & Conflict-Resolution Policy

- **Priority Rank:** Resume (1st) > LinkedIn > ATS JSON > Recruiter CSV > GitHub (5th)
- **Scalars (name, role):** Highest-priority source wins.
- **Arrays (emails, phones, skills):** Deduplicated UNION across all sources.
- **Conflict Detection:** Word-intersection name matching ensures rows belong to the same person.
- **Confidence:** Weighted average of all field confidences based on source priority.

4. Runtime Custom-Output Config (Projection)

The ProjectionEngine accepts a runtime JSON config to reshape the canonical record into any output schema — **zero backend code changes required.**

- **from:** Maps and renames fields (e.g., "emails[0]" → "primary_email").
- **include_provenance:** Boolean to toggle the provenance block.
- **on_missing:** Policy for missing data (null, omit, or error).
- **Validation:** Every projected output is validated via strict Pydantic v2 models before serving.

5. Edge Cases Handled vs Descoped

Handled:

- **LinkedIn profile ID parsed as phone:** URLs are actively stripped before phone regex runs.
- **Filename Job Titles:** "Preethi_AI_RESUME.pdf" extracts "Preethi AI" — domain keywords blacklisted.
- **Cross-Candidate Bleed:** Strict name matching prevents merging "Preethi S" with "Naveen V".
- **OCR Zero-Width Spaces:** Invisible "" artifacts breaking regex are stripped at ingestion.

Descoped (Under time pressure):

- **Scanned image PDFs:** No text layer; flagged gracefully but OCR (Tesseract) not implemented.
- **Live GitHub API rate limiting:** Stub implementation only; real-time scraping descoped to prioritize core merging logic.